

The OSGS residual projection: constrained versus unconstrained finite-element spaces, and why the constrained choice costs optimality

Notes for the manuscript

July 27, 2026

Abstract

The orthogonal-subgrid-scale (OSGS) method requires projecting the strong residual of the momentum–mass system onto a finite-element space. *Which* space? The unknown U_h lives in the Dirichlet-*constrained* product space $\mathcal{X}_{h0} = \mathcal{V}_{h0} \times \mathcal{Q}_{h0}$, so it is natural, and analytically convenient, to project the residual onto that same $\mathcal{X}_{h0}$. This note first builds that natural choice up sympathetically—following Codina’s own Remark 1, projecting onto the constrained space *removes* an analytical hypothesis and lets “all the results carry over”—and then explains, from first principles, exactly why the manuscript nonetheless projects onto the *unconstrained* spaces $\mathcal{X}_h = \mathcal{V}_h \times \mathcal{Q}_h$, on which $\boldsymbol{\pi}_h$ is free to be nonzero on $\partial\Omega$. The culprit is a one-element boundary strip on which the constrained projection is pinned to zero and cannot represent an $O(1)$ residual, seeding a spurious boundary layer. We teach the background as we go—the VMS scale split, the two subscale models, and L^2 -projection approximation theory—and we pay careful, honest attention to a subtlety that is easy to get wrong: precisely *how* optimality is affected. Codina *asserts* (Remark 1) that the constrained projection keeps *global* convergence optimal for general degree- k interpolation (only local boundary layers appear); the manuscript’s footnote makes the stronger, rate-level claim that it “spoils the optimal $O(h^{k+1})$ convergence.” We reconcile the two, labelling clearly what is asserted, what is in genuine tension for $k \geq 2$, and what is empirical conjecture.

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1 The setting

1.1 The continuous porous problem

We consider steady incompressible flow through an inhomogeneous porous medium on a bounded domain $\Omega \subset \mathbb{R}^d$. Writing $\mathbf{u}$ for the velocity, p for the pressure, ν for the kinematic viscosity, $\alpha = \alpha(\mathbf{x})$ for the (spatially varying) porosity, and $\boldsymbol{\sigma} = \boldsymbol{\sigma}(\alpha, \mathbf{u})$ for the Darcy–Brinkman–Forchheimer resistance tensor, the strong form of the momentum and mass balances reads, schematically,

$$\alpha \mathbf{a} \cdot \nabla \mathbf{u} - \nabla \cdot \left(2\alpha\nu \overset{\text{DS}}{\Pi} \nabla^s \mathbf{u} \right) + \boldsymbol{\sigma} \mathbf{u} + \alpha \nabla p = \mathbf{f} \quad \text{in } \Omega, \quad (1a)$$

$$\nabla \cdot (\alpha \mathbf{u}) = 0 \quad \text{in } \Omega, \quad (1b)$$

with $\mathbf{u} = \mathbf{0}$ on the Dirichlet boundary Γ_D (the analysis and the numerical experiments take $\Gamma_N = \emptyset$). Here $\mathbf{a}$ is the advective velocity; in the linearised (Oseen-type) analysis it is a prescribed field with $\nabla \cdot (\alpha \mathbf{a}) = 0$. This is exactly the manuscript’s strong momentum and mass equations (paper, eq:StrongMomentumEquation) and (paper, eq:StrongMassEquation); the resistance $\boldsymbol{\sigma}$ is symmetric positive semi-definite, of Forchheimer form $\boldsymbol{\sigma} = a(\alpha) + b(\alpha)|\mathbf{u}|$ (paper, eq:DBFResistanceTerm). The two dimensionless groups that organise the regimes are the Reynolds number Re and the Darcy number Da .

It is convenient to package eq. (1) as a single differential operator acting on the pair $U := [\mathbf{u}; p]$. Let $\mathcal{L}_{\mathbf{a}}$ be the first-plus-higher-order differential operator of the system and F the data, so that eq. (1) reads $\mathcal{L}_{\mathbf{a}} U = F$. The associated *strong residual* of an approximation U_h is

$$\mathcal{R}U_h := F - \mathcal{L}_{\mathbf{a}} U_h, \quad (2)$$

the pointwise amount by which U_h fails to satisfy the strong equations. The residual is the single object around which everything below turns. Its *first-order* part—the part that dominates the stabilisation—is

$$\mathbf{z} := \alpha(\mathbf{a} \cdot \nabla \mathbf{u} + \nabla p), \quad \delta := \nabla \cdot (\alpha \mathbf{u}), \quad (3)$$

the convective–pressure momentum residual and the mass residual. The reaction $\boldsymbol{\sigma} \mathbf{u}$ and the viscous term are, respectively, a zeroth-order and a second-order contribution; we return to the role of $\boldsymbol{\sigma}$ in section 7.

1.2 The Galerkin spaces: constrained versus unconstrained

The weak form is posed on $\mathcal{X}_0 := \mathcal{V}_0 \times \mathcal{Q}_0$, where $\mathcal{V}_0 := \{\mathbf{v} \in H^1(\Omega)^d : \mathbf{v}|_{\Gamma_D} = \mathbf{0}\}$ (this reduces to $H_0^1(\Omega)^d$ when $\Gamma_N = \emptyset$) and $\mathcal{Q}_0 := L^2(\Omega)$ (the zero-mean subspace in the all-Dirichlet case); see (paper, eq:WeakForm, §2). The subscript 0 records the essential point: *the trace vanishes on Γ_D* .

Discretising with continuous Lagrange elements of degree k_u for velocity and k_p for pressure ($k_p \leq k_u + 1$), we obtain *two* families of spaces, and the whole note lives in the gap between them.

Definition 1 (Constrained and unconstrained FE spaces). Let $\mathcal{V}_h \subset H^1(\Omega)^d$ and $\mathcal{Q}_h \subset H^1(\Omega) \cap C^0(\bar{\Omega})$ be the *unconstrained* degree- (k_u, k_p) finite-element spaces—no boundary condition is imposed. Their Dirichlet-*constrained* subspaces are

$$\mathcal{V}_{h0} := \mathcal{V}_h \cap H_0^1(\Omega)^d, \quad \mathcal{Q}_{h0} := \mathcal{Q}_h \cap \mathcal{Q}_0,$$

i.e. the functions in $\mathcal{V}_h, \mathcal{Q}_h$ that vanish on Γ_D . Write the product spaces $\mathcal{X}_h := \mathcal{V}_h \times \mathcal{Q}_h$ (unconstrained) and $\mathcal{X}_{h0} := \mathcal{V}_{h0} \times \mathcal{Q}_{h0}$ (constrained).

This matches the manuscript verbatim: the constrained $\mathcal{X}_{h0} := \mathcal{V}_{h0} \times \mathcal{Q}_{h0}$ appears at (paper, §3.2) and the appendix’s line “ $\mathcal{V}_{h0} := \mathcal{V}_h \cap H_0^1(\Omega)^d$, $\mathcal{Q}_{h0} := \mathcal{Q}_h \cap \mathcal{Q}_0$, $\mathcal{X}_{h0} := \mathcal{V}_{h0} \times \mathcal{Q}_{h0}$ ” (paper, App., oa: setup). It is also exactly Codina’s distinction between $V_h \subset V = H^1(\Omega)$ and its subspace $V_{h,0}$ [1].

Remark (Where does the discrete solution live?). *Always in the constrained space.* The Galerkin trial/test functions must honour $\mathbf{u}_h|_{\Gamma_D} = \mathbf{0}$, so the discrete unknown is $U_h \in \mathcal{X}_{h0}$, and the Galerkin problem tests against $V_h \in \mathcal{X}_{h0}$. This never changes. What is in question is a *different* object—the space onto which the residual is *projected*—and the temptation of this note is precisely the (mistaken) reflex “the residual should be projected where the unknown lives.”

2 Subscales and the two stabilizations

2.1 The variational multiscale scale split

Standard Galerkin fails for this problem twice over: it cannot control convection at high Re, and equal-order $\mathbf{u}/p$ interpolation violates the inf-sup condition. The Variational Multiscale (VMS) framework [5] repairs both from a single idea; the stabilised method used here generalises the stationary formulation of [3] and its transient orthogonal-subscale counterpart [2] to an inhomogeneous porous medium. Decompose the continuous solution space as a direct sum of what the mesh can resolve and what it cannot,

$$\mathcal{X}_0 = \mathcal{X}_{h0} \oplus \tilde{\mathcal{X}}_0, \quad \text{so that} \quad U = U_h + \tilde{U}, \quad U_h \in \mathcal{X}_{h0}, \quad \tilde{U} \in \tilde{\mathcal{X}}_0, \quad (4)$$

where $\tilde{\mathcal{X}}_0$ is the (infinite-dimensional) *subscale* or subgrid-scale space, the complement of the finite-element space (paper, §3.1, *Scale Splitting*; the displayed splitting there is unlabelled). The exact solution equals the resolved part U_h plus a subscale $\tilde{U}$ that lives below the mesh. Testing the full weak form against resolved test functions $V_h \in \mathcal{X}_{h0}$ gives the *resolved-scale equation* (paper, eq:WeakFormResolvedScales); the subscale $\tilde{U}$ appears there, and the game is to model it and eliminate it.

2.2 The exact subscale equation and its closure

Restricting the subscale equation to an element K and using the VMS modelling assumptions (M1) the subscale vanishes on element boundaries, (M2) inter-element contributions are negligible, and (M3) the operator inverse is well approximated by a matrix of *stabilisation parameters*, $\mathcal{L}_a^{-1}|_K \approx \tau_K$, the subscale equation collapses to a projection statement (paper, `eq:weak_form_subscales_as_projection`),

$$\tilde{\Pi}[\mathcal{L}\tilde{U}] = \tilde{\Pi}[\mathcal{R}U_h], \quad (5)$$

in which $\tilde{\Pi}$ projects onto the subscale space. Solving elementwise (paper, `eq:simplified_weak_form_subscales`),

$$\boxed{\tilde{U}|_K \approx \tau_K \tilde{\Pi}[\mathcal{R}U_h]|_K} \quad (6)$$

The subscale is the (τ -weighted) *projection of the strong residual*. Feeding eq. (6) back into the resolved equation yields the stabilised finite-element method (paper, `eq:VMSWeakFormSystem`). Everything hinges on the operator $\tilde{\Pi}$: it decides which part of the residual is fed back as a subscale.

2.3 ASGS versus OSGS: two choices of $\tilde{\Pi}$

There are two canonical closures.

ASGS (algebraic subgrid scales). Take the subscale space to be the space of finite-element residuals themselves; then $\tilde{\Pi} = \mathcal{I}$, the identity, and the *whole* residual is retained, $\tilde{U}|_K = \tau_K \mathcal{R}U_h|_K$. There is no projection to speak of.

OSGS (orthogonal subgrid scales). Take the subscale space to be the L^2 -orthogonal complement of the finite-element space, $\mathcal{X} = \mathcal{X}_{h0}^\perp$. Then $\tilde{\Pi}$ is the *orthogonal projection off* the finite-element space,

$$\tilde{\Pi} = \mathcal{I} - \Pi_{\tau h} \quad \Longleftrightarrow \quad \tilde{U}|_K = \tau_K (\mathcal{R}U_h - \pi_h)|_K, \quad \langle W_h, \pi_h \rangle = \langle W_h, \mathcal{R}U_h \rangle, \quad (7)$$

where $\Pi_{\tau h}$ is the (τ -weighted, in practice L^2) projection onto the finite-element space and $\pi_h = \Pi_{\tau h}[\mathcal{R}U_h]$ is the projected residual. The retained subscale is the *fluctuation* $\tilde{\Pi}[\mathcal{R}U_h] = (\mathcal{I} - \Pi_{\tau h})[\mathcal{R}U_h]$: only the residual's component *orthogonal to the mesh* survives. This is (paper, `eq:OSGSProjection`) with the τ -inner product (paper, `eq:TauInnerProduct`), simplified to plain L^2 (paper, `eq:L2InnerProduct`; this simplification and its stabilising properties are due to [2]). The full discrete system is (paper, `eq:OSGSProblem`): find $U_h \in \mathcal{X}_{h0}$ with

$$\begin{aligned} B_S(\mathbf{u}; U_h, V_h) &= L_S(\mathbf{u}, V_h, \pi_h), \\ \tilde{U}|_K &= \tau_K(\mathbf{u})(\mathcal{R}U_h - \pi_h)|_K \quad \forall K, \\ \langle W_h, \pi_h \rangle &= \langle W_h, \mathcal{R}U_h \rangle, \\ U &= U_h + \tilde{U}, \end{aligned}$$

and ASGS is recovered simply by setting $\pi_h = \mathbf{0}$.

Intuition. The projection question exists *only for OSGS*. In ASGS there is no π_h at all: $\tilde{\Pi} = \mathcal{I}$ retains the raw residual and there is nothing to project. The entire discussion that follows—onto *which* finite-element space is π_h computed—is an OSGS-only question. We will not refer to ASGS again except to say this.

Notice what eq. (7) demands of us: a projection operator $\Pi_{\tau h}$ needs a target space. The unknown U_h lives in $\mathcal{X}_{h0}$; the test functions W_h defining π_h range over ... what? That is the one degree of freedom left, and it is the subject of the note.

3 The natural first choice: project onto $\mathcal{X}_{h0}$

Let us make the constrained choice look as attractive as it genuinely is, because it is the choice one reaches for first, and for good reasons.

Reason 1: the projected residual then lives with the unknown. If W_h ranges over the constrained $\mathcal{X}_{h0}$, then $\pi_h = \Pi_{\tau_h}[\mathcal{R}U_h] \in \mathcal{X}_{h0}$ lives in the very same space as U_h . The subscale $\tilde{U}$, the resolved field U_h , and the projected residual π_h are then all objects of one type—elements of a Dirichlet-vanishing space. There is an appealing tidiness to a method in which every finite-dimensional quantity honours the same boundary condition.

Reason 2: the analysis is strictly simpler. This is not a matter of taste; it is a theorem-count. In the OSGS stability proof one needs a special test function built from the projection of the first-order residual onto the *constrained* velocity space, $\Pi_{1,0}z_h$ (it must be constrained to be an admissible, boundary-vanishing test function). If the *method's* projector is *also* the constrained one, then the two computable pieces of the residual—the fluctuation $\Pi_1^\perp z_h$ controlled by the Galerkin diagonal, and $\Pi_{1,0}z_h$ supplied by the test function—are the two complementary halves of *one and the same orthogonal projection*. They add up, by Pythagoras, to the full norm $\|z_h\|_{\tau_1}^2$ exactly, with constant $\beta_0 = 1$. No extra hypothesis is needed to recover control of the full residual.

Reason 3: Codina says the results carry over. This is stated, precisely, in [1, Remark 1]. In that work the OSGS (there “OSS”) methods are defined by projecting the residual onto the unconstrained W_h ; Remark 1 records what happens if one uses the constrained $W_{h,0}$ instead. Verbatim:

Both methods I and II could be slightly modified by projecting onto $W_{h,0}$ in (18) instead of projecting onto W_h . This would simplify the analysis presented in the following section, since the stability condition (35) stated there would not be needed, and all the results to be presented carry over to this case. However, even though the global convergence is optimal, projecting onto $W_{h,0}$ leads to spurious numerical boundary layers, similar to those found for the pressure in classical fractional step schemes for the transient problem (see for example [25]). Further discussion about this point can be found in [18].

Read the first two sentences on their own and the constrained choice looks like a free lunch: it *removes* a hypothesis (condition (35), which we meet in section 5) and *all the results carry over*. If a modelling choice buys a simpler, less-assumption-heavy theory at no cost to convergence, why would anyone do otherwise?

The answer is in the third sentence, and it is the whole point of this note.

4 Why the constrained projection is abandoned: the boundary strip

4.1 A refresher on L^2 -projection approximation theory

Let $\Pi_h : L^2(\Omega) \rightarrow \mathcal{V}_h$ be the L^2 projection onto the *unconstrained* degree- k space: $\Pi_h g$ is the unique $g_h \in \mathcal{V}_h$ with $(g - \Pi_h g, v_h) = 0$ for all $v_h \in \mathcal{V}_h$. Because it is an orthogonal projection it is, by construction, the *best* L^2 -approximation,

$$\|g - \Pi_h g\|_{L^2(\Omega)} = \inf_{v_h \in \mathcal{V}_h} \|g - v_h\|_{L^2(\Omega)}. \quad (8)$$

Choosing v_h to be a nodal interpolant of a smooth $g \in H^{k+1}$ and invoking Bramble–Hilbert gives the optimal fluctuation estimate

$$\|g - \Pi_h g\|_{L^2(\Omega)} \leq C h^{k+1} |g|_{H^{k+1}(\Omega)}. \quad (9)$$

The estimate eq. (9) is derived on the unconstrained space: the competitor v_h in eq. (8) is allowed to match g up to interpolation error *everywhere*, including at boundary nodes. This freedom is exactly what will be revoked in the constrained case.

In the OSGS method the retained subscale is precisely this kind of fluctuation, $\tilde{\Pi}[\mathcal{R}U_h] = \Pi_1^\perp[\mathcal{R}U_h]$ with $\Pi_1^\perp := \mathcal{I} - \Pi_1$, but projected in the $\boldsymbol{\tau}$ -weighted broken norm rather than in plain L^2 . One caveat keeps the bookkeeping honest. The first-order residual $\mathbf{z} = \alpha(\mathbf{a} \cdot \nabla \mathbf{u} + \nabla p)$ carries *one derivative fewer* than $\mathbf{u}$ (and the pressure lives only in H^k), so a plain L^2 best-approximation of $\mathbf{z}$ on its own would give merely $O(h^{k-1})$, *not* the $O(h^{k+1})$ one might read off eq. (9). The missing two orders are supplied by the $\boldsymbol{\tau}$ -weighting: on the momentum residual $\tau_1^{1/2} \sim h$, and the paper’s *weighted* best-approximation lemma (paper, App., oa:lem:bestapprox, oa:eq:WeightedApprox), whose $h_K^{2r} \tau_{i,K}$ weighting supplies the extra order, is what makes the consistency estimate close at the optimal rate. In other words the plain- L^2 bound eq. (9) is the *intuition*, and the weighted lemma is the *theorem*; what matters here—and what the constrained projection will destroy—is the same qualitative fact in either norm: away from the boundary the fluctuation is an *interpolation-error-sized* object.

4.2 The pinned strip

Now switch the projector’s target to the constrained $\mathcal{V}_{h0}$. Let $\Pi_{1,0}$ be the L^2 projection onto $\mathcal{V}_{h0}$. Every admissible competitor $v_h \in \mathcal{V}_{h0}$ satisfies $v_h = \mathbf{0}$ on Γ_D ; the projection is *pinned to zero on the boundary*. But the field being projected does not vanish there.

Proposition 1 (Annihilation of the local projection on the strip). *Let $\mathbf{z} = \alpha(\mathbf{a} \cdot \nabla \mathbf{u} + \nabla p)$ be the exact first-order momentum residual eq. (3), and suppose it is smooth enough to have a well-defined trace on Γ_D with $\mathbf{z}|_{\Gamma_D} \neq \mathbf{0}$ (the generic case). Let Ω_h denote the one-element strip of elements touching Γ_D , so $|\Omega_h| = O(h)$ on a shape-regular mesh. Then the constrained fluctuation satisfies, pointwise,*

$$\Pi_{1,0}^\perp \mathbf{z} = \mathbf{z} - \Pi_{1,0} \mathbf{z} = O(1) \quad \text{on } \Omega_h,$$

and consequently, in the plain L^2 norm,

$$\left\| \Pi_{1,0}^\perp \mathbf{z} \right\|_{L^2(\Omega)} \geq \left\| \Pi_{1,0}^\perp \mathbf{z} \right\|_{L^2(\Omega_h)} \approx O(1) \cdot |\Omega_h|^{1/2} = O(h^{1/2}), \quad (10)$$

in place of the $O(h^{k+1})$ of eq. (9).

Heuristic. On Γ_D every $v_h \in \mathcal{V}_{h0}$ is zero, so the pointwise defect there is forced to equal the trace itself, $(\mathbf{z} - v_h)|_{\Gamma_D} = \mathbf{z}|_{\Gamma_D} = O(1)$, uniformly in h : best approximation eq. (8) cannot remove an error the constraint hard-wires in. A continuous Π_k function that is $O(1)$ on Γ_D but returns toward the interior best approximation must carry that $O(1)$ magnitude across the one boundary layer of elements, a band of thickness $O(h)$ along a boundary of $(d-1)$ -measure $O(1)$. Hence $\int_{\Omega_h} |\Pi_{1,0}^\perp \mathbf{z}|^2 \approx O(1)^2 \cdot O(h)$, which is eq. (10). The exponent $\frac{1}{2}$ comes entirely from the $O(h)$ *measure* of the strip, not from any approximation power; that is why it does not improve with k . $\square$

That $\mathbf{z}|_{\Gamma_D} \neq \mathbf{0}$ is not exotic: on Γ_D the velocity is Dirichlet-fixed, yet the momentum equation forces $\mathbf{a} \cdot \nabla \mathbf{u} + \nabla p = \alpha^{-1}(\mathbf{f} + \nabla \cdot (2\alpha\nu \overset{\text{DS}}{\Pi} \nabla^s \mathbf{u}) - \boldsymbol{\sigma} \mathbf{u})$, which has no reason to vanish there. So a constrained projection of the residual is generically annihilated on a strip where the true residual is $O(1)$: *the local projection is destroyed exactly where the subscale is supposed to live.*

4.3 The fractional-step analogy

This mechanism is not new; it is the spatial twin of a well-known defect in time-splitting methods, and Codina’s Remark 1 makes the analogy explicitly (“similar to those found for the pressure in classical fractional step schemes”). In a fractional-step / pressure-projection scheme the velocity–pressure decoupling silently over-imposes a boundary condition—effectively a spurious homogeneous Neumann condition $\partial p / \partial n = 0$ that the true pressure does not satisfy—and the mismatch concentrates in a thin wall layer of width $O(\sqrt{\nu \Delta t})$, the *artificial pressure boundary layer*, which caps pressure accuracy [7, 6]. In both cases an approximation or decomposition step forces an incorrect boundary behaviour and the error localises in an $O(\text{width})$ strip along Γ where mesh (or time-step) refinement of the interior cannot reach it. The constrained OSGS projection manufactures precisely the spatial version of that layer.

5 Exactly how optimality is affected — a careful reconciliation

Here is the pedagogical payoff, and the one place where it is easy to overclaim. There are *two* statements in the literature about the consequence of the constrained projection, and they are *not* the same statement. We put them side by side, honestly.

5.1 (a) What Codina 2008 asserts (Remark 1): optimal global rate, local layers

Codina’s Remark 1 is a *qualitative-defect* statement, not a rate-loss statement—and it is a *remark*, asserting that “all the results ... carry over” to the constrained variant without giving a separate proof for it. Re-read the key clause, verbatim:

... even though the global convergence is optimal, projecting onto $W_{h,0}$ leads to spurious numerical boundary layers ...

Global convergence *remains optimal*; the constrained projection produces *spurious boundary layers*, a local, qualitative artefact. The rate in the relevant (mesh-dependent, $\boldsymbol{\tau}$ -weighted energy) norm of the resolved field is not claimed to degrade. The manuscript’s own appendix transcribes this faithfully: it states only that the constrained choice “would simplify the analysis (assumption H:projection would not be needed) but produces spurious numerical boundary layers” (paper, App.), and its remark on the analysed-versus-implemented formulation asserts no rate change (paper, App., oa:rem:analyzed).

5.2 (b) The power-counting puzzle, and its resolution

Proposition 1 produced an $O(1)$ defect on an $O(h)$ -measure strip, i.e. $O(h^{1/2})$ in the plain L^2 norm. For $k \geq 1$ we have $h^{1/2} \gg h^{k+1}$ as $h \rightarrow 0$; taken at face value this would wreck an $O(h^{k+1})$ target. How, then, can Codina keep the *global* rate optimal? Two independent mechanisms, both real, resolve the puzzle; we flag which parts are firm and which are conjecture.

(i) **The layer lives in the subscale, not the resolved field.** The $O(1)$ object is $\tilde{\Pi}[\mathcal{R}U_h] = \Pi_{1,0}^\perp[\mathcal{R}U_h]$ —the *fluctuation*, hence the subscale $\tilde{U}$ and the reconstructed total field $U = U_h + \tilde{U}$. Codina’s “optimal global convergence” is a statement about the *resolved* U_h ; the spurious layer is a feature of the subscale/total-field reconstruction, precisely the analogue of the fractional-step pressure layer he cites. The resolved velocity is not itself made to carry the $O(1)$. [*Firm as an attribution of Codina’s claim; the extent of leakage into U_h is the open point of part (c).*]

(ii) **The working norm is τ -weighted, and the strip is mesh-thin.** The plain- L^2 count is simply the *wrong norm* for Codina’s theorem: convergence is measured in the τ -weighted broken norm. Redo eq. (10) with the weight. On Ω_h ,

$$\|\Pi_{1,0}^\perp[\alpha X]\|_{\tau_1}^2 = \sum_{K \subset \Omega_h} \tau_{1,K} \cdot O(1) \cdot |K| \approx \tau_1 \cdot O(1) \cdot h \implies \|\cdot\|_{\tau_1} \sim (\tau_1 h)^{1/2}. \quad (11)$$

In Codina’s diffusion-dominated Oseen setting ($\sigma = 0$), $\tau_1 \sim h^2/(c_1 \alpha \nu)$, so $(\tau_1 h)^{1/2} \sim h^{3/2}/\nu^{1/2}$. Now, Codina asserts optimal global convergence for *general* degree- k interpolation—his hypothesis H6 takes $\mathbf{u} \in H^{k+1}$, $p \in H^k$ with $k \geq 1$, and Remark 1 says of the constrained variant that “all the results ... carry over”; he nowhere confines optimality to $k = 1$. For $k = 1$ the optimal energy rate is $O(h)$ and the strip contribution $h^{3/2}$ is *higher order*, hence invisible in the global rate: optimality is uncontroversial. For $k \geq 2$, however, the optimal energy rate is $h^k \geq h^2$ while this crude $O(1)$ -on-the-strip count still gives $h^{3/2}$, which *would* then dominate—a genuine tension with Codina’s general- k Remark 1. [*We flag this and do not pretend to resolve it. Because Remark 1 is an informal remark with no detailed proof for the constrained variant, we cannot tell whether the crude $O(1)$ amplitude overestimates the true boundary defect (so that optimality survives for $k \geq 2$ after all) or whether the “carry-over” claim is optimistic there. What is firm is the $k = 1$ case, the existence of the tension for $k \geq 2$, and the fact that the manuscript sidesteps the whole question by projecting onto the unconstrained space. We do not read Codina as having restricted optimality to $k = 1$.*]

5.3 (c) The manuscript’s stronger claim, and where it rests

The manuscript’s footnote to (paper, eq:OSGSPProblem) says something one notch stronger. Verbatim:

In our implementation the projection (eq:NonlinearResidualProjection) is computed with W_h ranging over the velocity and pressure finite element spaces without their Dirichlet constraints, so that π_h is not forced to vanish on $\partial\Omega$. Projecting on the constrained space $\mathcal{X}_{h0}$ instead would force the boundary degrees of freedom to vanish (the space $\mathcal{X}_{h0}$ carrying homogeneous Dirichlet conditions), annihilating the local L^2 projection there and introducing an $O(1)$ boundary residual that spoils the optimal $O(h^{k+1})$ convergence.

The phrase “spoils the optimal $O(h^{k+1})$ convergence” is a *rate* claim. It goes beyond Codina’s “optimal global convergence with boundary layers,” and—read literally as a statement about the global rate of the resolved field in the energy norm—it is in tension with the paper’s *own* appendix, which claims only spurious layers. This is the single sentence in the cluster that must be read carefully.

The honest reading is that the footnote’s rate-level claim is *not* Codina’s theorem and is *not* proved in the appendix (which explicitly declines to analyse the constrained variant). It rests

instead on the manuscript’s *own manufactured-solution (MMS) evidence* in the porous/reaction setting, and it can be true *simultaneously* with Codina’s theorem because the two speak about different norms and different regimes:

- *Different norm.* The MMS convergence gate measures the plain- L^2 velocity and continuity error of the *resolved* field, via the scale-free indicators $\varepsilon_M, \varepsilon_C$. Plain L^2 is exactly the norm in which the $O(1)$ -on- $O(h)$ count *does* bite ($O(h^{1/2})$)—if the spurious stabilisation forcing contaminates U_h . Codina’s theorem uses the τ -weighted energy norm of eq. (11), which damps the strip.
- *Different regime.* Codina is $\sigma = 0$ Oseen; the manuscript is porous, with reaction σ and varying porosity α absent from his setting, and it probes high Da. At high Da the exact solution develops a Brinkman screening layer of width $\ell_\sigma = (\alpha\nu/\sigma)^{1/2}$; when h under-resolves ℓ_σ , the true first-order residual $\mathbf{z}$ is already steep and large near Γ_D , and the constrained projection—still pinned to zero—superimposes its spurious layer on a genuine, under-resolved physical one, so the effective “ $O(1)$ ” is $O(\text{large})$. It is plausible that this sharpens the qualitative layer into a measurable rate loss. [Conjecture: the paper does not analyse the constrained variant; and note two honest counter-currents—(1) for constant σ the reaction residual has no fluctuation at all (section 7), so σ never drives the layer directly; (2) high Da makes $\tau_1 \approx 1/\sigma$ small, which suppresses the velocity fluctuation in the energy norm. The rate reading therefore rests on physical-layer amplification plus plain- L^2 contamination of U_h , neither of which is established here.]
- *Possibly $k \geq 2$.* As the power count in (b)(ii) shows, degradation is far more available for $k \geq 2$ than for the $k = 1$ case Codina certifies.

The disciplined conclusion: keep the footnote’s rate phrasing as an *empirical, regime-specific* claim about the plain- L^2 MMS gate, and do not present it as Codina’s theorem or as a proven global-rate result. Codina’s qualitative statement and the paper’s empirical extension are compatible, not contradictory.

5.4 (d) The price of the honest choice: condition (35)

If the constrained projection is so simple analytically, what does the *unconstrained* projection—the one actually implemented and analysed—cost? Exactly one hypothesis. In the stability proof the Galerkin diagonal controls the fluctuation $\Pi_1^\perp \mathbf{z}_h$ (now off the *unconstrained* space), while the special test function still supplies $\Pi_{1,0} \mathbf{z}_h$ (necessarily off the *constrained* space, to be admissible). These two are projections onto *different* spaces, so they no longer sum to $\|\mathbf{z}_h\|_{\tau_1}^2$ by Pythagoras. The missing glue is the manuscript’s assumption `H:projection` (paper, `eq:ProjectionStability`), verbatim:

Projection stability. *Let Π_1 and $\Pi_{1,0}$ be the τ_1 -weighted L^2 projections of a velocity field onto the unconstrained and the constrained velocity spaces, and $\Pi_1^\perp := I - \Pi_1$. There is $\beta_0 > 0$, independent of h , such that for all $V_h = [\mathbf{v}_h; q_h] \in \mathcal{X}_h$, writing $\mathbf{z}_h := \alpha(\mathbf{a} \cdot \nabla \mathbf{v}_h + \nabla q_h)$,*

$$\|\tau_1^{1/2} \mathbf{z}_h\|_h \leq \frac{1}{\beta_0} \left(\|\tau_1^{1/2} \Pi_{1,0} \mathbf{z}_h\|_h + \|\tau_1^{1/2} \Pi_1^\perp \mathbf{z}_h\|_h \right). \quad (12)$$

This is exactly the Codina–Blasco-type inf–sup condition (35) of [1] (the equal-order-interpolation device it descends from originates in [4]), which in his own notation reads

$$\|z_h\|_{\tau_1} \leq \frac{1}{\beta_0} (\|\Pi_{\tau_1,0}(z_h)\| + \|\Pi_{\tau_1}^\perp(z_h)\|_{\tau}), \quad z_h \equiv \mathbf{a} \cdot \nabla \mathbf{v}_h + \nabla q_h.$$

It says precisely: a bound on the constrained-projection part $\Pi_{1,0}z_h$ and on the unconstrained fluctuation $\Pi_1^\perp z_h$ suffices to bound the *whole* residual $\|z_h\|_{\tau_1}$ —i.e. the component of z_h that lies in $\mathcal{V}_h$ but is τ_1 -orthogonal to $\mathcal{V}_{h0}$ is not independent of the other two. (For the divergence part it holds trivially with $\beta_0 = 1$, because $\mathbf{v}_h = \mathbf{0}$ on $\partial\Omega$ forces $\nabla \cdot \mathbf{v}_h$ to have zero mean.) It is needed precisely because B_S is *not* coercive; eq. (12) is what recovers inf–sup stability from the two controllable projections.

Constrained projection: easy analysis, no eq. (12), but boundary layers.

Unconstrained projection: needs eq. (12), but clean boundaries — and it is the honest method that is implemented.

6 Settling the question: a degree-dependent theorem

We can now do more than flag the tension of section 5: in the OSGS energy norm we can *prove* the answer, and it turns out to depend on the polynomial degree. Write $X(V) := \mathbf{a} \cdot \nabla \mathbf{v} + \nabla q$ for the first-order operator, so that $\mathbf{z} = \alpha X(U)$ is the momentum residual of eq. (3), and let

$$\eta_\Pi := \|\Pi^\perp[\alpha X(U)]\|_{\tau_1} \quad (13)$$

be the *consistency defect* of the method whose residual projector is Π (either Π_1 , the unconstrained projection, or $\Pi_{1,0}$, the constrained one). Everything below is set in the fixed-data, viscosity-dominated asymptotic regime that governs the convergence *rate* (section 5): $\tau_1 = \Theta(h^2)$ (so $\tau_1^{1/2} = \Theta(h)$) and $\tau_2 = \Theta(1)$ (from $\tau_2 \rightarrow \nu/\alpha$; hence, by the design bound $\varepsilon\tau_2 \leq C_2$, $\varepsilon = O(1)$ —a fact we will need). Throughout, $||| \cdot |||_O$ is the projection-free triple norm (paper, `oa:eq:TripleNorm`), *the same* for both variants; only the method’s S_1 changes.

6.1 The question reduces to the size of one number

Lemma 1 (Reduction to the consistency defect). *Let B_{OSGS} be OSGS with projector Π , inf-sup stable in $||| \cdot |||_O$ with an h -independent constant (for the constrained variant this is Codina’s Remark 1). Then*

$$|||U - U_h|||_O \leq C(\Psi_O(h) + \eta_\Pi) = C(\Theta(h^k) + \eta_\Pi), \quad (14)$$

where $\Psi_O(h) = \Theta(h^k)$ is the interpolation error in the energy norm ($k := k_u$).

Proof. The analyzed OSGS method is not strongly consistent; by the appendix’s consistency identity (paper, `oa:lem:consistency`), for all $W_h \in \mathcal{X}_{h0}$,

$$B_{\text{OSGS}}(\mathbf{a}; U - U_h, W_h) = S_1(U, W_h) + S_2(U, W_h), \quad S_1(U, W) = (\Pi^\perp[\alpha X(U)], \alpha X(W))_{\tau_1}.$$

Let $\widehat{U}_h \in \mathcal{X}_{h0}$ be the nodal interpolant of U and $\mathbf{e}_h := U_h - \widehat{U}_h$. Inf-sup stability gives a W_h with $|||W_h|||_O = 1$ and

$$\beta |||\mathbf{e}_h|||_O \leq B_{\text{OSGS}}(\mathbf{e}_h, W_h) = \underbrace{B_{\text{OSGS}}(U - \widehat{U}_h, W_h)}_{\text{(I)}} - \underbrace{(S_1(U, W_h) + S_2(U, W_h))}_{\text{(II)}},$$

using $B_{\text{OSGS}}(U_h, W_h) = L(W_h)$ and the identity above. We do *not* invoke a general continuity of B_{OSGS} (it fails with an h -independent constant, the convective term $(\mathbf{w}, \alpha \mathbf{a} \cdot \nabla \mathbf{u})$ carrying an h^{-1}). Instead, term (I) is bounded by the appendix's interpolation lemma (paper, `oa:lem:interpolation`), valid precisely because $U - \hat{U}_h$ is a *small, boundary-vanishing* interpolation error: $|(I)| \leq C \Psi_O(h) \|W_h\|_O$. Term (II): by Cauchy–Schwarz in $(\cdot, \cdot)_{\tau_1}$ and the full-residual term of the triple norm, $\|\alpha X(W_h)\|_{\tau_1} \leq \|W_h\|_O$, so $|S_1(U, W_h)| \leq \eta_{\Pi} \|W_h\|_O$; the mass part S_2 is $O(h^{k_p+1})$ (the pressure carries no Dirichlet data, $\mathcal{Q}_{h0}$ being only zero-mean, so it is unaffected by the constraint). Collecting and adding $\|U - \hat{U}_h\|_O = \Theta(h^k)$ gives eq. (14). $\square$

So the whole question is: *how big is η_{Π} ?*

6.2 A sharp two-sided bound on the constrained defect

Lemma 2 (The constrained defect is $\Theta(h^{3/2})$, the unconstrained one $O(h^{k+1})$). *Assume $\mathbf{z}|_{\Gamma_D} = \alpha X(U)|_{\Gamma_D} \neq \mathbf{0}$ on a subset of Γ_D of positive $(d-1)$ -measure (the generic case). Then, on a shape-regular quasi-uniform family in the viscous regime,*

$$\eta_0 := \|\Pi_{1,0}^\perp[\alpha X(U)]\|_{\tau_1} = \Theta(h^{3/2}), \quad \text{while} \quad \eta_1 := \|\Pi_1^\perp[\alpha X(U)]\|_{\tau_1} = O(h^{k+1}). \quad (15)$$

The exponent $3/2$ is independent of k ; only the hidden constant $c_k > 0$ decreases (mildly) with k .

Proof. Upper bound. By the minimizing property of the τ_1 -orthogonal projection (paper, `oa:eq:BestApprox`), $\eta_0 \leq \|\alpha X(U) - w_h\|_{\tau_1}$ for every $w_h \in \mathcal{V}_{h0}$. Take w_h the *constrained* Scott–Zhang interpolant, which reproduces the homogeneous nodal boundary values. Split the mesh into the interior and the one-element boundary strip $\Omega_h := \bigcup \{K : \bar{K} \cap \Gamma_D \neq \emptyset\}$, $|\Omega_h| = O(h)$. On interior elements the constraint is inactive and the standard estimate gives $\sum_K \tau_{1,K} \alpha_K^2 h_K^{2k} \|\cdot\|_{H^k(K)}^2 \lesssim (h^2/\nu) h^{2k} = h^{2k+2}/\nu$. On a strip element K the competitor must vanish on the face $F = \bar{K} \cap \Gamma_D$ while $\alpha X(U)|_F \neq \mathbf{0}$; no polynomial vanishing on F can match a nonzero trace, so $\|\alpha X(U) - w_h\|_{L^2(K)}^2 \geq c_k |K|$ with $c_k > 0$ (a subspace-obstruction / Markov fact: a fixed-degree polynomial cannot form a boundary layer thinner than $O(h)$). Summing over the $O(h^{-(d-1)})$ strip elements, each weighted by $\tau_{1,K} = \Theta(h^2/\nu)$,

$$\sum_{K \subset \Omega_h} \tau_{1,K} \|\alpha X(U) - w_h\|_{L^2(K)}^2 \lesssim \frac{h^2}{\nu} \cdot O(1) \cdot \underbrace{h^d \cdot h^{-(d-1)}}_{|\Omega_h| = O(h)} = \frac{h^3}{\nu}.$$

Since $2k + 2 > 3$ for every $k \geq 1$, the strip dominates and $\eta_0 \leq C h^{3/2} \nu^{-1/2}$, with *no* k in the exponent. The same computation with the *unconstrained* Scott–Zhang interpolant (which matches the trace) has no strip penalty and reproduces the appendix's weighted best approximation (paper, `oa:lem:bestapprox`), giving $\eta_1 = O(h^{k+1})$.

Lower bound (the strip penalty is real). It suffices to exhibit one configuration. On a uniform 1D mesh of $(0, 1)$, size h , with linear elements, τ_1 is elementwise *constant* (so the τ_1 -weighted projection is the plain L^2 projection) *and* of magnitude $\tau_1 = \Theta(h^2)$. Project $g \equiv 1$ (smooth, $g|_{\partial\Omega} = 1 \neq 0$) onto $\mathcal{V}_{h0}$. With the interior hat basis the normal equations read $\frac{1}{6}c_{i-1} + \frac{2}{3}c_i + \frac{1}{6}c_{i+1} = 1$; the constant $c_i \equiv 1$ solves every interior row *exactly*, so the nodal error $e_i := 1 - c_i$ satisfies the homogeneous recurrence $e_{i-1} + 4e_i + e_{i+1} = 0$ with $e_0 = e_n = 1$. Its characteristic roots are $-2 \pm \sqrt{3}$; the decaying mode has ratio $2 - \sqrt{3} \approx 0.27$, so the error is $\Theta(1)$ on $O(1)$ elements (width $O(h)$) and geometrically negligible inside. On the first element $\int_0^h (g - \Pi_{1,0}g)^2 = \frac{h}{3}(1 - (2 - \sqrt{3}) + (2 - \sqrt{3})^2) = \Theta(h)$, whence $\|g - \Pi_{1,0}g\|_{L^2} = \Theta(h^{1/2})$ and, since this projection is the exact minimizer, nothing is smaller. Therefore $\eta_0 = \tau_1^{1/2} \|g - \Pi_{1,0}g\|_{L^2} = \Theta(h) \cdot \Theta(h^{1/2}) = \Theta(h^{3/2})$. The same one-element layer along a flat boundary patch, with $O(h^{-(d-1)})$ faces and $|\Omega_h| = O(h)$, reproduces $\Theta(h^{3/2})$ in d dimensions. $\square$

Raising k helps only in the interior; it *cannot* repair the trace obstruction on the boundary, where a function pinned to $\mathbf{0}$ is asked to represent an $O(1)$ field. That single asymmetry between $\mathcal{V}_h$ and $\mathcal{V}_{h0}$ is the entire effect.

6.3 The degree-dependent verdict

Proposition 2 ($k = 1$: optimality is preserved). *For equal-order linear elements ($k = 1$) the constrained-projection OSGS retains the optimal energy rate, $|||U - U_h|||_O = \Theta(h)$.*

Proof. Lemmas 1 and 2 give $|||U - U_h|||_O \leq C(h + \eta_0) = C(h + \Theta(h^{3/2}))$. As $3/2 > 1$, $\eta_0 = \Theta(h^{3/2}) = o(h)$ sits *strictly beneath* the interpolation floor $\Theta(h)$ and is asymptotically invisible. Hence $|||U - U_h|||_O = \Theta(h)$. $\square$

The negative half needs one natural hypothesis. In the linearized (Oseen / Newton) problem the advecting field $\mathbf{a}$ is a velocity iterate, so it is legitimate—though not automatic in the appendix—to assume

$$\mathbf{a} = \mathbf{0} \text{ on } \Gamma_D \quad (\text{the advecting field obeys no-slip}). \quad (\text{H}_\Gamma)$$

Under (H_Γ) , on Γ_D the boundary residual is a *gradient*, $\mathbf{z}|_{\Gamma_D} = \alpha \nabla p|_{\Gamma_D}$ (the convective part vanishes), and on the strip interior $\alpha \mathbf{a} \cdot \nabla \mathbf{u} = O(h)$ is subdominant. This is what lets a curl-free test field align with $\mathbf{z}$.

Proposition 3 ($k \geq 2$: optimality is spoiled, with a matching lower bound). *Assume (H_Γ) , the genericity of lemma 2, and the viscous regime ($\tau_1 = \Theta(h^2)$, $\tau_2 = \Theta(1)$, hence $\varepsilon = O(1)$). For $k \geq 2$,*

$$|||U - U_h|||_O = \Theta(h^{3/2}) < \Theta(h^k), \quad (16)$$

so the constrained projection genuinely degrades the energy rate to $3/2$.

Proof. The upper bound is lemmas 1 and 2: $|||U - U_h|||_O \leq C(h^k + \eta_0) = O(h^{3/2})$ since $3/2 < k$. For the matching lower bound, use a *pressure* test function $V^* = [\mathbf{0}; q^*]$ with $q^* \in \mathcal{Q}_{h0}$ a strip-localized finite-element bump chosen so that $\alpha \nabla q^* = \mathbf{z}$ on Ω_h up to lower order—admissible because, by (H_Γ) , $\mathbf{z}$ is a gradient there, and because the pressure is *not* Dirichlet-constrained (so q^* is free on Γ_D).

Norm of the vehicle. With $\mathbf{v}^* = \mathbf{0}$ the viscous, reaction and mass terms of $|||V^*|||_O$ vanish, leaving $\varepsilon \|q^*\|^2 = O(1) \cdot h^3 = O(h^3)$ (here $\varepsilon = O(1)$ is essential; a gradient- $O(1)$ bump of width h has amplitude $O(h)$, so $\|q^*\|^2 \sim h^3$) and $\|\alpha X(V^*)\|_{\tau_1}^2 = \|\alpha \nabla q^*\|_{\tau_1}^2 \sim \tau_1 \cdot O(1) \cdot |\Omega_h| = h^2 \cdot h = h^3$. Thus $|||V^*|||_O = \Theta(h^{3/2})$.

The pairing does not cancel. By idempotent self-adjointness of $\Pi_{1,0}^\perp$ and $\alpha \nabla q^* = \mathbf{z}$ on Ω_h ,

$$S_1^0(U, V^*) = (\Pi_{1,0}^\perp \mathbf{z}, \alpha \nabla q^*)_{\tau_1} = (\Pi_{1,0}^\perp \mathbf{z}, \mathbf{z})_{\tau_1} + (\text{l.o.}) = \|\Pi_{1,0}^\perp \mathbf{z}\|_{\tau_1}^2 + (\text{l.o.}) = \eta_0^2 = \Theta(h^3),$$

the lower-order correction controlled by $\eta_0 \cdot o(h^{3/2}) = o(h^3)$ because $\Pi_{1,0}^\perp \mathbf{z}$ is strip-concentrated (the exponential profile of lemma 2); and $S_2(U, V^*) = 0$.

Restricted continuity, then the bound. Because $\mathbf{v}^* = \mathbf{0}$, the non-continuous convective term $(\mathbf{v}, \alpha \mathbf{a} \cdot \nabla \mathbf{u})$ is absent, and one checks termwise—using $\tau_2 = \Theta(1)$ for the divergence pairing $||[\tau_2^{-1} q^*]| \leq (\alpha/\nu)^{1/2} \|q^*\| \sim h^{3/2}$ —that $|B_{\text{OSGS}}^0(W, V^*)| \leq C |||W|||_O |||V^*|||_O$ for *this* pair. Expanding as in lemma 1 with $\mathbf{e}_h = U_h - \hat{U}_h$,

$$B_{\text{OSGS}}^0(\mathbf{e}_h, V^*) = \underbrace{B_{\text{OSGS}}^0(U - \hat{U}_h, V^*)}_{\leq C h^k \cdot h^{3/2} = h^{k+3/2}} - \underbrace{S_1^0(U, V^*)}_{\Theta(h^3)} = -\Theta(h^3) \quad (k \geq 2, \text{ since } h^{k+3/2} = o(h^3)).$$

Restricted continuity then gives $|||e_h|||_0 \geq |B_{\text{OSGS}}^0(e_h, V^*)|/(C|||V^*|||_0) = \Theta(h^3)/\Theta(h^{3/2}) = \Theta(h^{3/2})$, and $|||U - U_h|||_0 \geq |||e_h|||_0 - |||U - \hat{U}_h|||_0 \geq c h^{3/2} - C h^k = \Theta(h^{3/2})$ for $k \geq 2$. With the upper bound this is two-sided. $\square$

Intuition. The boundary blemish has a *fixed* size $\Theta(h^{3/2})$, set by the geometry of the $O(h)$ strip, not by the elements; the target it must beat is the shrinking h^k . For $k = 1$ the target h is coarser than $h^{3/2}$, so the blemish hides beneath it; for $k \geq 2$ the target h^k is finer than $h^{3/2}$, so the blemish sticks out and *becomes* the rate. Optimality survives iff $3/2 > k$, i.e. iff $k = 1$.

6.4 The tension, resolved

Propositions 2 and 3 settle section 5 in the energy norm: Codina’s Remark 1—“all the results carry over”—is *exactly right for $k = 1$ and false for $k \geq 2$* . There is no contradiction with anything Codina proves: his consistency estimate ([1, Lemma 3]) is derived for the *unconstrained* method via a best-approximation competitor that *matches the boundary trace*—the very step that fails on $\mathcal{V}_{h0}$ —and his optimal Q_1/Q_2 numerics are that unconstrained method; the constrained variant is supported only by the informal Remark 1 and is never analysed or tested for $k \geq 2$. Symmetrically, the manuscript’s footnote is right for the higher-order elements ($\mathbb{P}_2$) it actually uses.

Two honest caveats bound the claim. First, the regime is load-bearing: the argument uses $\tau_2 = \Theta(1)$ (hence $\varepsilon = O(1)$); were τ_2 to scale like h^2 the pressure vehicle would fail, so the verdict is tied to the viscous-dominated asymptotics that fix the rate. Second, the theorem is in the *energy* norm. The manufactured-solution gate and the footnote’s “ $O(h^{k+1})$ ” measure the plain- L^2 velocity error, and there the story is subtler: the defect enters the duality (Aubin–Nitsche) estimate *quadratically* ($\eta_0 \cdot \eta_0^{\text{dual}} = \Theta(h^3)$), so clean theory would leave the $k = 1$ L^2 rate optimal and only partly degrade $k \geq 2$. We do not bank on that recovery—the duality for an inconsistent, non-coercive form is exactly the open route of the appendix, and this codebase’s own MMS evidence reports that the constrained projection breaks the $O(h^{k+1})$ gate in practice. The energy-norm statement above is the part that is *proved*; the plain- L^2 verdict for $k \geq 2$ is decided cleanly by a single numerical experiment—project onto $\mathcal{V}_{h0}$ and measure the $\mathbb{P}_2$ rate—which we recommend as the definitive check.

7 A caution: two different “annihilations”

The word *annihilation* appears twice in this cluster with completely unrelated meanings, and conflating them would be a serious error.

The GOOD annihilation (intended). The OSGS appendix’s “Exact annihilations” lemma (paper, App., oa:lem:annihilation) states, for constant σ and ε and all $U_h, V_h \in \mathcal{X}_{h0}$:

$$\Pi_1^\perp(\sigma \mathbf{u}_h) = \mathbf{0} \text{ and } \Pi_2^\perp(\varepsilon p_h) = 0 \text{ (reactive and compressibility parts of the residual have no fluctuation); } \dots (\Pi_1^\perp z, \sigma \mathbf{v}_h)_{\tau_1} = 0 \text{ and } (\Pi_2^\perp w, \varepsilon q_h)_{\tau_2} = 0 \text{ for any } z, w \dots$$

The mechanism: $\sigma \mathbf{u}_h \in \mathcal{V}_h$ already, and the *unconstrained* fluctuation $\Pi_1^\perp = \mathcal{I} - \Pi_1$ annihilates *anything already in the projection space $\mathcal{V}_h$* . This is a desirable structural fact—the reactive and compressibility residuals carry no subscale, so OSGS keeps the full $\|\sigma^{1/2} \mathbf{v}\|$ norm and needs no reaction renormalisation. *Constancy of σ is essential*: a variable σ would carry $\sigma \mathbf{u}_h$ out of $\mathcal{V}_h$.

The BAD annihilation (this note). Proposition 1’s “annihilation of the local projection” is the destruction of $\Pi_{1,0}\mathbf{z}$ on the boundary strip because *constrained* functions must vanish there. It has nothing to do with residual components lying in a space; it is a defect *caused by* the Dirichlet constraint on the projection target.

Same word, opposite valence. The good annihilation is $\Pi_1^\perp$ killing what is in $\mathcal{V}_h$; the bad annihilation is the constraint killing $\Pi_{1,0}$ on Γ_D . Never conflate them.

8 Summary

	Constrained proj. onto $\mathcal{X}_{h0}$	Unconstrained proj. onto $\mathcal{X}_h$
Projected residual π_h lives in	$\mathcal{X}_{h0}$ (vanishes on Γ_D)	$\mathcal{X}_h$ (free on Γ_D)
Condition eq. (12) / (35)	not needed ($\beta_0 = 1$, Pythagoras)	required (Codina–Blasco inf-sup)
Boundary behaviour of the subscale	$O(1)$ on the $O(h)$ strip (pinned)	$O(h^{k+1})$ interpolation fluctuation
Codina 2008 verdict (Remark 1, asserted)	optimal <i>global</i> rate + spurious <i>local</i> boundary layers (general k ; but see the $k \geq 2$ tension, §5(b))	optimal, clean boundaries
This paper’s claimed verdict (empirical/conjectural)	rate-level $O(h^{k+1})$ <i>claimed</i> spoiled in the porous/high-Da plain- L^2 MMS gate — regime-specific, not proved for the constrained variant	optimal $O(h^{k+1})$ MMS convergence
Implemented & analysed here?	no	yes

The constrained projection is the natural first reach and the easier theory—it even removes hypothesis eq. (12). It is abandoned for one reason: on the one-element strip touching Γ_D it is pinned to zero and cannot represent the $O(1)$ first-order residual, annihilating the local projection and seeding a spurious boundary layer (the spatial twin of the fractional-step pressure layer). Codina *asserts* (Remark 1) that this keeps the *global* rate optimal for general k while polluting the boundary (with a genuine, unresolved tension at $k \geq 2$, §5(b)); the manuscript’s stronger, rate-level phrasing is an empirical extension resting on its own porous MMS, and is best read as a claim about the plain- L^2 gate rather than as Codina’s energy-norm theorem. The price of the honest unconstrained projection—and it is a fair price—is the projection-stability condition eq. (12), which glues the constrained projection and the unconstrained fluctuation back into control of the full residual.

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